

Solutions to the Practice Midterm

1. True or False

(a)

If A is a 4×3 matrix then the system $AX = b$ must be inconsistent for some $b \in \mathbb{R}^4$.

True

A 4×3 matrix has at most 3 pivot columns, so it cannot have a pivot in every row. Therefore the columns of A cannot span $\mathbb{R}^4$. Hence there exists some $b \in \mathbb{R}^4$ such that $AX = b$ is inconsistent.

(b)

If A is a 3×4 matrix then $AX = 0$ must have infinitely many solutions in $\mathbb{R}^4$.

True

There are 4 variables but at most 3 pivots, so there is at least one free variable. Therefore the homogeneous system has infinitely many solutions.

(c)

If A is a 10×5 matrix then the system $AX = 0$ is consistent.

True

Every homogeneous system has the trivial solution $X = 0$.

(d)

If m vectors $v_1, \dots, v_m$ span $\mathbb{R}^4$ and are linearly independent, then $m = 4$.

True

A basis for $\mathbb{R}^4$ must contain exactly 4 vectors.

(e)

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be linear. Then T is one-to-one if every column of the standard matrix has a pivot position.

True

A pivot in every column means there are no free variables in $AX = 0$, so the only solution is the trivial solution. Hence T is one-to-one.

(f)

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ 1 \\ 3 \end{bmatrix} \right\}$$

is linearly independent.

False

Any set containing the zero vector is linearly dependent.

(g)

A set of 5 vectors in $\mathbb{R}^4$ is always linearly dependent.

True

There cannot be more than 4 linearly independent vectors in $\mathbb{R}^4$.

(h)

$$\left\{ \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \begin{bmatrix} 2 \\ 8 \end{bmatrix} \right\}$$

is linearly independent.

False

Since

$$\begin{bmatrix} 2 \\ 8 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 4 \end{bmatrix},$$

the vectors are linearly dependent.

(i)

$$\begin{bmatrix} 1 & 5 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

is invertible.

False

The matrix has a zero row, so it does not have 3 pivots.

(j)

If a 3×3 matrix has 3 pivot positions, then it is invertible.

True

A square matrix is invertible exactly when it has a pivot in every row and every column.

(k)

A system $Ax = b$ is inconsistent if $[A|b]$ has a pivot in the last column.

True

That produces a row of the form

$$[0 \ 0 \ \cdots \ 0 \mid 1],$$

which corresponds to the impossible equation $0 = 1$.

(l)

Every homogeneous system $Ax = 0$ has at least one solution.

True

The trivial solution $x = 0$ always works.

(m)

If A is a 4×3 matrix, the columns of A can span $\mathbb{R}^4$.

False

There are only 3 columns, so at most 3 pivot columns. Spanning $\mathbb{R}^4$ requires 4 pivots.

(n)

The column space of a 5×3 matrix is a subspace of $\mathbb{R}^3$.

False

Each column has 5 entries, so the column space is a subspace of $\mathbb{R}^5$.

2. Solve the system

$$\begin{cases} x_1 + 2x_2 + x_3 + 3x_4 = 1 \\ 2x_1 + 4x_2 + 3x_3 + 8x_4 = 4 \\ 3x_1 + 6x_2 + 4x_3 + 11x_4 = 5 \end{cases}$$

Augmented matrix:

$$\left[\begin{array}{cccc|c} 1 & 2 & 1 & 3 & 1 \\ 2 & 4 & 3 & 8 & 4 \\ 3 & 6 & 4 & 11 & 5 \end{array} \right]$$

$$R_2 \rightarrow R_2 - 2R_1, \quad R_3 \rightarrow R_3 - 3R_1$$

$$\left[\begin{array}{cccc|c} 1 & 2 & 1 & 3 & 1 \\ 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 2 & 2 \end{array} \right]$$

$$R_3 \rightarrow R_3 - R_2$$

$$\left[\begin{array}{cccc|c} 1 & 2 & 1 & 3 & 1 \\ 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

From row 2:

$$x_3 + 2x_4 = 2$$

$$x_3 = 2 - 2x_4$$

From row 1:

$$x_1 + 2x_2 + x_3 + 3x_4 = 1$$

Substitute $x_3 = 2 - 2x_4$:

$$x_1 + 2x_2 + 2 - 2x_4 + 3x_4 = 1$$

$$x_1 = -1 - 2x_2 - x_4$$

Let

$$x_2 = s, \quad x_4 = t.$$

Then the Parametric Vector Form is given by:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 2 \\ 0 \end{bmatrix} + s \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

3. Find A^{-1}

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 3 \\ 0 & 1 & 1 \end{bmatrix}$$

Start with

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & -1 & -2 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$R_3 \rightarrow R_3 - R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & -1 & -2 & 1 & 0 \\ 0 & 0 & 2 & 2 & -1 & 1 \end{array} \right]$$

$$R_3 \rightarrow \frac{1}{2}R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & -1 & -2 & 1 & 0 \\ 0 & 0 & 1 & 1 & -\frac{1}{2} & \frac{1}{2} \end{array} \right]$$

$$R_1 \rightarrow R_1 - 2R_3, \quad R_2 \rightarrow R_2 + R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 1 & -1 \\ 0 & 1 & 0 & -1 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & 1 & -\frac{1}{2} & \frac{1}{2} \end{array} \right]$$

Hence

$$A^{-1} = \begin{bmatrix} -1 & 1 & -1 \\ -1 & \frac{1}{2} & \frac{1}{2} \\ 1 & -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

Check that it is correct by showing $AA^{-1} = I$.

4.

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \end{bmatrix}$$

(a) Solve $AX = 0$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1, \quad R_3 \rightarrow R_3 - 3R_1$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 3 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_2 + 2x_3 + 3x_4 = 0$$

$$x_2 = -2x_3 - 3x_4$$

$$x_1 + x_2 + x_3 + x_4 = 0$$

$$x_1 = x_3 + 2x_4$$

Let

$$x_3 = s, \quad x_4 = t.$$

Then

$$X = s \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 2 \\ -3 \\ 0 \\ 1 \end{bmatrix}$$

(b) Solve $AX = \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix}$

A particular solution is

$$X_p = e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

since

$$Ae_3 = \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix}.$$

Therefore

$$X = X_p + X_h$$

and hence

$$X = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 2 \\ -3 \\ 0 \\ 1 \end{bmatrix}$$

(c)

Is $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ onto?

No

The matrix has only 2 pivots, so the columns do not span $\mathbb{R}^3$.

5. Geometric interpretation

$$T(X) = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} X$$

Apply T to the standard basis vectors:

$$T(e_1) = \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \quad T(e_2) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Hence T is a rotation by 90° clockwise.

Rotation by $\frac{\pi}{2}$ clockwise

6. Find the standard matrix

Rotation by $5\pi/3$:

$$R = \begin{bmatrix} \cos \frac{5\pi}{3} & -\sin \frac{5\pi}{3} \\ \sin \frac{5\pi}{3} & \cos \frac{5\pi}{3} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$$

Reflection about $x_2 = 0$:

$$F = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Since reflection happens after rotation,

$$T = FR$$

$$T = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$$

$$T = \begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix}$$

7. Show that T is not linear

$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2^x + 2^y - 2 \\ x + y \\ x - y \end{bmatrix}$$

A linear transformation must satisfy

$$T(cv) = cT(v).$$

for all scalars c and all vectors v . But

$$T \left(2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) = T \left(\begin{bmatrix} 2 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix}$$

and

$$2T \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) = 2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$$

Therefore

T is not linear

8.

Given

$$T \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad T \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

First write the standard basis vectors in terms of the given vectors.

$$e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Thus, because T is linear:

$$T(e_1) = \frac{1}{2} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1/2 \\ 0 \\ 0 \end{bmatrix}$$

Similarly,

$$e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Hence

$$T(e_2) = \begin{bmatrix} 1/2 \\ -1/2 \\ 0 \\ 0 \end{bmatrix}$$

Therefore the standard matrix is

$$A = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Now compute

$$T \begin{bmatrix} 4 \\ 6 \end{bmatrix} = A \begin{bmatrix} 4 \\ 6 \end{bmatrix} = \begin{bmatrix} 5 \\ -1 \\ 0 \\ 0 \end{bmatrix}$$

$$T \begin{bmatrix} 4 \\ 6 \end{bmatrix} = \begin{bmatrix} 5 \\ -1 \\ 0 \\ 0 \end{bmatrix}$$